

Category	Tr Part	Risk	Likelihood (1=Unlikely, 5 = Certain)	nificant, 5=catastrophic)	Priority	Tr Impact	Proposed action
Electronic	Battery	Battery connector disconnects under vibration	2	5 High	High	Immediate power loss ends the run	Use a locking connector and strain relief
Electronic	Battery	Battery capacity lower than expected on competition day	2	4 High	High	The robot cannot sustain the full mission duration	Budget energy conservatively and verify with full-run tests
Electronic	Battery	Voltage sag under simultaneous motor and conveyor load	3	4 High	High	Controllers brown out during peak demand	Measure peak current and add supply margin
Electronic	Power distribution	Reverse power connection	1	5 Medium	Medium	Electrical components can be damaged	Include reverse-polarity protection and use non-reversible connectors such as XT30 or XT60
Electronic	Power distribution	Protections preventing regeneration current of motors from going back to the battery	1	5 Medium	Medium	Electrical components can be damaged	Avoid reverse-current protections that block motor regeneration back to the battery
Electronic	Power distribution	Excessive inrush current because of excessive capacitance load on the power path	1	5 Low	Low	Electrical components, mainly motors, can be damaged	Include a precharge circuit on the robot
Electronic	Power distribution	Brownout when multiple actuators start together	3	4 High	High	Compute and sensors reboot in motion	Sequence high-current starts and add bulk capacitance
Electronic	Power distribution	Fuse nuisance trip during acceleration	2	3 Medium	Medium	The robot shuts down even without a real fault	Tune fuse ratings to measured transient currents
Electronic	Power distribution	Ground reference shift between sensors and compute	2	3 Medium	Medium	Measurements become biased or noisy	Adopt a controlled grounding scheme
Electronic	Power distribution	Power rail noise resets the camera or USB devices	2	4 High	High	Perception disappears intermittently during motion	Filter the rail and isolate noisy loads
Electronic	Motors	Motor driver thermal protection engages on the ramp	2	4 High	High	Traction is lost in the most demanding zone	Heat-sink the driver and validate worst-case current
Electronic	Motors	Encoder cable break or intermittent contact	2	4 High	High	Odometry quality collapses unexpectedly	Use flexible strain-relieved cabling
Electronic	Sensors	USB cable disconnect on the camera or LIDAR	2	4 High	High	Critical perception input disappears mid-run	Secure cables mechanically and minimize free length
Electronic	Sensors	Sensor startup surge overloads the regulator	1	4 Medium	Medium	Boot sequence fails intermittently	Stagger sensor startup or oversize the regulator
Electronic	Sensors	EMI noise from motors affecting sensors	3	3 High	High	Localization and perception measurements become noisy	Use twisted or shielded cables and add filtering
Electronic	Computation	MicroSD corruption after abrupt power loss	2	4 High	High	The robot may fail to boot for the next test or run	Use orderly shutdown during development and keep backups
Electronic	Computation	Cooling fan power loss on the compute board	2	3 Medium	Medium	Thermal margin disappears during long runs	Monitor fan operation and add passive cooling margin
Electronic	Computation	CPU overheating	2	3 Medium	Medium	Compute performance drops or the controller shuts down	Add a heat sink and verify thermal margins under load
Electronic	Wiring	Harness routed too close to moving mechanisms	2	4 High	High	Cables can be cut, pinched, or unplugged	Separate wiring from moving parts and protect crossings
Electronic	Wiring	Insulation abrasion on chassis edge	2	5 High	High	A short circuit can damage the platform	Add grommets and abrasion protection on edges
Electronic	Wiring	Connector polarity mismatch between similar plugs	2	4 High	High	Subsystem damage occurs during maintenance	Key connectors by voltage and label every branch
Electronic	Wiring	Shared return path overheats under peak current	1	4 Medium	Medium	Voltage drop and local damage appear in the harness	Size conductors from measured worst-case current
Electronic	Wiring	Loose connectors	3	4 High	High	Intermittent shutdowns or sensor dropouts occur during runs	Use locking connectors and strain relief on harnesses
Electronic	Safety	No accessible master power cutoff during tests	2	4 High	High	Faults are harder to contain during debugging	Add an external cutoff accessible from the chassis exterior
Electronic	Safety	Unprotected battery terminals during handling	1	5 High	High	Accidental short can cause fire or injury	Use insulated covers and a handling protocol
Mechanical	Robot Body	Getting stuck on the top of the ramp	4	5 High	High	Robot cannot enter or leave Zone 4	Increase ground clearance and validate breakover angle on the ramp
Mechanical	Robot Body	Ground clearance insufficient at door threshold	2	4 High	High	The robot becomes stuck before entering a zone	Chamfer the lower frame and validate clearance on mockups
Mechanical	Robot Body	Fasteners loosening due to repeated shocks	3	3 High	High	Subsystem alignment degrades over time	Use locking nuts and a pre-run tightening checklist
Mechanical	Robot Body	Robot tipping when hitting blocks or climbing the ramp	2	4 High	High	Sensor alignment is lost and the run may stop	Lower the center of mass and limit speed during impacts and slope traversal
Mechanical	Robot Body	Camera mount shifts after collision	2	3 Medium	Medium	Calibration is invalidated after an impact	Stiffen the mount and protect it from frontal hits
Mechanical	Robot Body	LIDAR mount misalignment after impact	2	4 High	High	Localization degrades abruptly mid-run	Use a rigid isolated mount and a reference alignment mark

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Mechanical	Robot Body	Structural resonance near motor operating speed	2	3 Medium	Sensors vibrate and positioning precision falls	Avoid resonant frequencies and add local stiffening
Mechanical	Chassis	Ramp slope too steep for stable ascent	2	4 High	The robot stalls or slides while entering Zone 4	Keep a low center of mass and reserve torque margin
Mechanical	Drive train	Ball caster vibration	3	2 Medium	Camera images degrade and detection becomes unreliable	Use damping or a larger smoother caster
Mechanical	Drive train	Wheel hub loosening	2	4 High	Drive accuracy drops and mechanical damage can occur	Use locking hardware and inspect hubs before each run
Mechanical	Drive train	Wheel diameter mismatch	3	2 Medium	Straight-line motion accumulates heading error	Measure the wheels and compensate in calibration
Mechanical	Drive train	Motor mount loosening under vibration	2	4 High	Drivetrain alignment shifts during the run	Reinforce mounts and use thread-locking fasteners
Mechanical	Drive train	Payload shifts center of mass on the ramp	2	3 Medium	Climbing stability decreases and traction is reduced	Keep storage low and centered within the chassis
Mechanical	Button mechanism	Door pusher too short to reach reliable contact	2	3 Medium	Button actuation fails despite correct navigation	Dimension the pusher with measured clearance margin
Mechanical	Button mechanism	Door pusher bends under load	2	3 Medium	Applied force is lost before actuating the button	Increase section stiffness or shorten the unsupported length
Mechanical	Button mechanism	Pusher snags on the door frame while retreating	2	4 High	Robot gets stuck at the zone entrance	Round the tip geometry and verify retreat clearance
Mechanical	Block Collection	Intake funnel width too narrow	2	3 Medium	Blocks are deflected away instead of guided inward	Tune the funnel opening from real block dimensions
Mechanical	Block Collection	Malfunction (blocks getting stuck or not going in)	2	3 High	Damage or blockage in the intake path	Dimension the fan-based intake carefully and test it early
Mechanical	Block Collection	Block jam inside storage container	3	3 High	Blocks cannot move toward the delivery mechanism	Add smooth internal guiding surfaces and avoid sharp corners
Mechanical	Block Collection	Intake funnel width too wide	2	2 Low	Two blocks enter together and jam the transfer path	Reduce the throat width and add separators
Mechanical	Block Collection	Fan intake leaks around the duct	2	3 Medium	Suction is too low to capture blocks reliably	Seal the duct joints and validate pressure loss
Mechanical	Block Collection	Sweeper geometry misses blocks at the edges	3	2 Medium	Collection rate falls on side-approach targets	Widen the effective sweep envelope
Mechanical	Block Collection	Collector contacts decorative elements	1	4 Medium	Arena elements or the robot can be damaged	Limit intake protrusion and validate approach envelopes
Mechanical	Block Collection	Multiple blocks entering intake simultaneously	3	3 High	The intake or storage entrance jams	Narrow the intake channel and add internal guides
Mechanical	Block Collection	Block stuck at intake entrance	3	3 High	Collection stops until the blockage is cleared	Provide agitation, vibration, or a self-clearing geometry
Mechanical	Block Collection	Block bouncing away from intake	3	2 Medium	The robot misses blocks that were initially reachable	Add a compliant ramp or softer contact surface at the intake
Mechanical	Storage	Storage volume deforms under full load	1	4 Medium	The delivery path changes geometry and jams approach	Reinforce panels and test with maximum payload
Mechanical	Storage	Block fragments accumulate inside storage	2	2 Low	Transfer friction rises over repeated cycles	Clean the storage path and avoid abrasive contacts
Mechanical	Storage	Capacity sensor mount loosens	2	3 Medium	Fullness indication becomes unreliable	Rigidly mount the sensor and protect the cable
Mechanical	Delivery	Delivery opening too narrow for clustered blocks	2	3 Medium	Blocks jam at the exit despite moving transport	Increase outlet width or guide blocks into single file
Mechanical	Delivery	Conveyor pulley misalignment	2	3 Medium	The belt tracks poorly and rubs against the frame	Use alignment slots and assembly jigs
Mechanical	Delivery	Blocks rebound back inside during release	2	2 Low	A partial payload remains after an apparently successful release	Shape the exit chute to favor one-way motion
Mechanical	Delivery	Delivery chute angle too shallow	2	2 Low	Blocks slow down and stop before leaving the robot	Increase chute slope and reduce surface friction
Mechanical	Delivery	Door or curtain does not reset after release	2	3 Medium	Next collection cycle begins with an open storage bin	Add passive return force and end-stop guidance
Mechanical	Delivery	Conveyor belt slipping	2	3 Medium	Blocks are transported unreliably during delivery	Add belt tension adjustment and a textured belt surface
Mechanical	Delivery	Loose belt	3	3 Medium	Efficiency reduction and unreliable delivery	Keep an adjustable position on the secondary pulley

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Mechanical	Delivery		Blocks not going out	3	5 High			Inability to release blocks and loss of the main func	Detect motor movement and verify that storage empties after a full turn
Mechanical	Delivery		Ball bearings failure	1	5 Medium			Permanent damage to the mechanism	Dimension the rotating elements carefully
Software	Movement		Wheel encoder failure	2	4 High			Loss of odometry and navigation accuracy	Monitor encoder values and switch to fused estimation when anomalies are detected
Software	Movement		Wheel traction difference	3	3 Medium			Robot drifts while moving straight and misses align	Calibrate wheel speeds and use closed-loop heading control
Software	Movement		Carpet traction variation in Zone 2	3	3 High			Motion model errors accumulate on the carpeted s	Tune separate drive parameters for carpet traversal
Software	Localization		Pose estimate jumps after loop closure	2	4 High			Path tracking becomes inconsistent after remappin	Gate loop-closure corrections and replan after large jumps
Software	Localization		Map corruption after repeated deliveries	2	3 Medium			The corner reference becomes unreliable	Reset transient obstacle layers and preserve fixed landmarks
Software	Localization		Starting pose initialized incorrectly	2	4 High			All subsequent targets are offset from the true arer	Add a startup localization verification step
Software	Localization		Wheel slip not detected by estimator	3	3 High			The fused pose slowly drifts away from reality	Compare commanded and observed motion to flag inconsistency
Software	Localization		LIDAR reflections on arena borders	2	3 Medium			False obstacles appear in the map	Filter implausible ranges and validate them against persistence
Software	Localization		Camera and LIDAR timestamps desynchronized	2	3 Medium			Sensor fusion uses inconsistent measurements	Synchronize clocks and reject stale data
Software	Localization		LIDAR occlusion	3	3 High			Map updates become incorrect and navigation dec	Mount the sensor higher and reject persistent occluded returns
Software	Localization		SLAM divergence	2	4 High			Robot loses the global reference and cannot reach	Reinitialize localization from known landmarks or arena features
Software	Localization		TF frame mismatch	2	4 High			Targets are transformed incorrectly and the robot d	Validate frame definitions and add runtime consistency checks
Software	Vision		Camera auto-exposure reacts too slowly	3	2 Medium			Detection quality drops after rapid heading change	Limit exposure adaptation range and pause briefly before capture
Software	Vision		Motion blur during approach	3	2 Medium			The target disappears in the final pickup phase	Reduce speed near blocks and use image sharpness checks
Software	Vision		Block occlusion by previously collected blocks	2	2 Low			Remaining blocks are underestimated in clustered	Revisit dense areas from multiple viewpoints
Software	Vision		Button not detected under oblique angle	2	4 High			Door actuation sequence cannot start	Scan the button area from a controlled viewing pose
Software	Vision		Full storage obstructs forward camera view	2	3 Medium			Perception quality degrades late in the mission	Keep sensors elevated and isolate storage from the optical path
Software	Vision		Camera lens obstruction	2	3 Medium			Detection range and confidence drop suddenly	Protect the lens and inspect it before every run
Software	Vision		Partial block visibility	3	2 Medium			Estimated block pose is wrong and pickup alignme	Require multi-frame confirmation before committing a pickup
Software	Navigation		Route to Zone 3 not replanned after failed button attempt	2	3 Medium			Robot wastes time repeating an impossible approa	Bound retries and fall back to a new alignment pose
Software	Navigation		Ramp approach angle chosen too aggressively	2	4 High			The robot clips the ramp edge and loses alignment	Constrain the entry pose and validate it before climbing
Software	Navigation		Recovery behavior loops indefinitely after obstacle encounter	2	4 High			The mission stalls even though hardware is healthy	Limit recovery retries and escalate to a different strategy
Software	Navigation		Goal queue not cleared after delivery	2	3 Medium			The robot chases stale block locations	Invalidate collected or delivered targets after each cycle
Software	Navigation		Planner ignores storage state during long collection pass	2	3 Medium			The robot keeps searching instead of delivering	Make storage state a hard planning constraint
Software	Navigation		Temporary obstacle layer never clears	2	2 Low			The arena appears blocked and the planner becom	Age out temporary obstacles automatically
Software	Navigation		Path planner oscillation	3	2 Medium			The robot replans repeatedly and wastes mission ti	Add hysteresis and commit windows for target selection
Software	Navigation		Return-to-corner trigger occurs too late	2	4 High			Payload overflows or blocks are lost before deliver	Use a conservative fullness threshold with hysteresis
Software	Navigation		Return-to-corner trigger occurs too early	3	2 Medium			The robot performs too many delivery cycles	Tune the trigger against measured storage capacity

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Software	Navigation	Deadlock in navigation loop	2	4 High	Robot stops operating while still powered	Add watchdog timers and a timeout-based recovery state
Software	Button interaction	Contact force command too high	2	3 Medium	The pusher or button support can be damaged	Limit actuation force and use compliant end geometry
Software	Button interaction	Door button misalignment	2	3 High	Zone 3 cannot be opened when needed	Use a staged alignment routine before contact
Software	Block Collection	Pickup sequence starts before final alignment converges	3	2 Medium	Blocks are pushed aside instead of collected	Gate the intake command on pose and heading tolerance
Software	Block Collection	Collector keeps running after intake is empty	2	2 Low	Power is wasted and wear increases	Stop the collector when transfer completion is detected
Software	Block Collection	Collected block not registered in internal state	2	3 Medium	The robot may overfill storage or revisit the same area	Cross-check intake events with storage observations
Software	Storage	Storage occupancy counter drifts from reality	3	2 Medium	Delivery decisions become inconsistent	Reconcile the counter with periodic sensor checks
Software	Storage	Incorrect storage capacity estimation	2	3 Medium	The robot keeps collecting when full and wastes time	Infer fill state from sensors or counted transfers
Software	Delivery	Delivery completion declared too early	2	4 High	Blocks remain inside and the next cycle starts with errors	Verify the empty state with sensors or timed transport margins
Software	Delivery	Delivery completion never declared	2	3 Medium	Robot waits at the corner and loses mission time	Add timeout-based exit criteria
Software	Computation	CPU load spike during simultaneous mapping and vision	3	3 High	Control latency grows and motion quality degrades	Separate critical loops and cap noncritical compute rates
Software	Computation	Node restart leaves stale state in the mission manager	2	3 Medium	Subsystem coordination becomes inconsistent after restart	Reinitialize shared state on restart
Software	Computation	Logging saturates storage bandwidth	2	2 Low	Real-time tasks suffer from I/O stalls	Throttle logging and disable verbose traces during competition
Software	Computation	Configuration file mismatch between subsystems	2	3 Medium	Controllers and planners run with inconsistent parameters	Version configuration files and validate them on startup
Software	Computation	Watchdog threshold set too aggressively	2	2 Low	Subsystems reset even under normal peak load	Tune watchdog timeouts from measured execution times
Software	Computation	Memory leak	2	3 Medium	Performance degrades over the duration of the run	Stress-test the code and monitor resource consumption
Software	Computation	Thread synchronization issues	2	4 High	State corruption or crashes can occur in operation	Protect shared data and keep interfaces deterministic